Exam 2: Study Guide

This study guide lists skills you need to be able to do for your exam. It does not tell you *how* to do the skills. Skills may be tested individually, together, or not at all. It is up to you to determine what skills you need already know, and which skills you need more practice on. Review relevant topics in ALEKS, problems from the end of each chapter of your textbook, worksheets, and lab experiments.

Chapter 4 – Compounds and Chemical Bonds

1) Classify properties as being physical or chemical properties.
   a) Section 4.1.1
2) Classify changes as being physical or chemical – and provide a statement to justify reasoning
   a) Section 4.1.2
3) Classify elements as being atomic elements or molecular elements
   a) Section 4.4
   b) (Know which 7 elements exist as diatomic molecules!)
4) Classify matter as pure substance (element, compound), or mixture (homogeneous, heterogeneous) – and provide a statement to justify reasoning.
   a) Section 4.2
5) Use the law of conservation of mass to calculate missing quantity
   a) Section 4.1.3
6) Determine the number of each type of atom in a chemical formula
   a) Section 4.3.1
7) Classify compounds as ionic (Type 1 and 2), acid (Binary and oxyacid), and molecular (covalent) based on chemical formula, name, or chemical model
   a) Section 4.4
8) Know common ion charges for Groups 1, 2, 15, 16 17.
   a) Section 4.5.1
9) Predict chemical formula for ionic compounds based on common ion charges
   a) Section 4.5.2
10) Name compounds that are molecular, ionic, and acids given chemical formula, or vice versa)
   a) Memorize your polyatomic ions!
   b) Section 4.6

Chapter 5 – The Mole and Chemical Formulas

1) Calculate the formula mass, molecular mass, or molar mass of a compound
   a) Section 5.3
2) Convert between units of mass (grams), units of quantity (moles), and counts (number of atoms, number of formula units, number of molecules)
   a) Section 5.2, 5.4, 5.5
3) Calculate the mass percent composition of a compound
   a) Section 5.6, 5.8
4) Identify how many moles of each element there are per mole of a compound.

a) Section 5.2

5) Give the empirical formula that corresponds to a molecular formula
   a) Section 5.7
6) Calculate empirical formula of a compound given experimental data, or mass percent composition of a compound.
   a) Section 5.7
7) Calculate the molecular formula of a molecular compound from empirical formula and molar mass.
   a) Section 5.9

Chapter 6 – Lewis Structures, Shapes, and Intermolecular Forces

1) Draw the Lewis dot structure for a single atom/ion, ionic compound, covalent (molecular) compound, or polyatomic ion.
   a) Section 6.1.1 (single atoms)
   b) Section 6.1.2 (ionic compounds)
   c) Section 6.1.3, 6.2 (molecular compounds)
2) Count the number of valence electrons that should appear in a Lewis dot structure.
   a) Section 6.2
3) Draw resonance structures for a compound/ion and Determine the number of resonance structures possible for a compound.
   a) Section 6.2.1
4) Count the number of electron groups around an atom.
   a) Section 6.3
5) Describe the bonding around an atom in terms of number of covalent bonds, bonding pairs (electrons), or lone pairs (electrons).
   a) Section 6.2, 6.3
6) Determine the electron group geometry or molecular geometry around an atom
   a) A single bond, a double bond, a triple bond, a lone pair each count as one electron group.
   b) Section 6.3
7) Determine angles (bond angles) between atoms in a molecule
   a) Section 6.4
8) Determine the type of bond that would form between two atoms given electronegativity values
   a) Section 6.4
      i) Pure Covalent (Non-polar covalent): <0.5
      ii) Polar Covalent: ≥ 0.5 and ≤ 2.0
      iii) Ionic: >2.0
9) Determine if a molecule is a polar molecule or non-polar molecule considering the types of covalent bonds present, and molecular geometry.
   a) Section 6.4
10) Be able to define intermolecular forces, and the different types of intermolecular forces.
   a) Differentiate between intermolecular forces and chemical bonds.
   b) Section 6.5
11) Predict what intermolecular forces would be present between two molecules given the molecular formula of a compound
   a) Section 6.5

b) You will need to be able to determine molecular geometry to predict if a molecule is polar
c) If question only gives one chemical formula, you are predicting forces between molecules of a sample of that pure compound.

Chapter 7 – Solids, Liquids and Phase Changes

1) Classify matter as a solid, liquid or a gas given information about its shape, volume, compressibility, and atomic/molecular motion or spacing – and provide a statement to justify reasoning.
   a) Section 7.1 (and introduction)
   b) Section 8.1
2) Convert between temperature scales (℃ , ℉ , K)
   a) Review from Chapter 1 (Measurements)
   b) Section 7.4.2
   c) Formulas will be given to you (like on your Yellow Sheet)
3) Define melting point, boiling point, specific heat capacity, enthalpy (of fusion, of vaporization).
   a) Section 7.5.1 (Specific Heat)
   b) Section 7.6 (Melting Point, Boiling Point)
4) Rank molecules in order of increasing (or decreasing) relative melting point, boiling point, viscosity, vapor pressure or surface tension based on the strength of the intermolecular forces between molecules of the same compound.
   a) Section 6.5
   b) Section 7.2
5) Know the names of physical changes involving states of matter (i.e. evaporation, condensation, boiling, freezing, melting, etc.) and recognize solids, liquids and gases based on drawings.
   a) Section 7.6.1, 7.6.2
6) Identify a chemical or physical change as being exothermic or endothermic – and provide a statement to justify reasoning.
   a) Section 7.4.1
7) Calculate heat, amount of substance (mass/mol), specific heat capacity or temperature change (provided the other values).
   a) Section 7.5.1, 7.5.2
   b) Constants (such as enthalpy ΔH and specific heat (c_p) will be provided in the question prompt)
   c) Unit conversions may be necessary
8) Be able to identify which states of matter would be present at a given temperature provided a heating curve, melting point and boiling point of a substance.
   a) Section 7.6.4
9) Calculate heat, amount of substance (mass/mol), or enthalpy (of fusion/vaporization) provided the other values.
   a) Section 7.6.3
10) Calculate the heat required to change the temperature of a substance when the temperature range involves a phase change.
   a) i.e. Calculate each segment and add together.
   b) (No examples of this in the book)
11) Classify solids based on type (ionic, molecular, metallic or network covalent)
   a) Section 7.3

b) The only network covalent compounds as far as you are concerned: ~~Graphite (C)~~, Diamond (C), and quartz (SiO_2)

This table will NOT be given to you. Memorize the names of shapes (geometries)!

Number of Electron Groups	Electron Group Geometry	Ideal Bond Angles	0 lone pairs	1 lone pair	2 lone pairs
			Molecular Geometry		
2	Linear	180°	Linear		
3	Trigonal Planar	120°	Trigonal Planar	Bent	
4	Tetrahedral	109.5°	Tetrahedral	Trigonal Pyramidal	Bent